



Verified, anonymous workplace reports on Midnight

Prove **that** you work there, never **who** you are.

The hotline dilemma

Anonymous channels

Drop boxes and open web forms get ignored or flooded. Nobody can tell an employee from an outsider, or one person from fifty.

Identified channels

HR email and named tickets carry weight, but people don't use them. Fear of retaliation is the top reason misconduct goes unreported.

What organisations actually need

Proof that the reporter is a real member, rate-limited and corroborated, with a guarantee that nobody (not the employer, not the vendor, not the chain) can tell which member.

Murmur: membership, not identity

Members only

ZK proof of a Merkle path to an enrolled commitment.

Nobody knows who

Secret, path and leaf stay in private state.

Selective disclosure

Reveal site, role, both or neither. Whatever is shown is proven true.

No flooding

One report per member per round via unlinkable nullifiers.

Anonymous corroboration

Once per member, and the circuit proves you're *not the author*.

Authorship, later

The reporter can prove a report is theirs to a key they choose, once.

Two ledgers, one proof

PRIVATE STATE · MEMBER'S DEVICE

secret
site, role
Merkle path to their leaf
never leaves the device

→
ZK proof
disclose()

PUBLIC LEDGER · MIDNIGHT

roster root & size
nullifiers (unlinkable)
report: category, severity, text hash, *site?* *role?*,
authorTag, corroborations, status

The proof says: *"I know a leaf in this roster; this nullifier is mine and fresh; the site/role I chose to show is the one I enrolled with."* Report text goes off-chain to the operator, and only its hash is on chain, so tampering shows.

The Compact contract

CIRCUIT	WHO	PROVES	ZKIR OPS	PROVER KEY
fileReport	member	membership · fresh round nullifier · honest disclosure	277	10.0 MB
corroborate	member	membership · fresh nullifier · not the author	315	10.0 MB
claimAuthorship	author	knows the secret behind authorTag	167	2.8 MB
enroll · advanceRound · setStatus	operator	knows the operator secret (hash sealed at deploy)	45–174	2.8 MB

compactc 0.35.0 · language 0.27 · runtime 0.20. `HistoricMerkleTree<10>`, `Set` nullifiers, `Map` reports, domain-separated `persistentHash`. Full ZK key generation in ~30 s.

Live demo: the real contract, in your browser

**Murmur**
Verified, anonymous workplace reports on Midnight

Compact contract running in your browser

No network · no wallet · no data leaves this tab

[▶ Guided story](#)

Reset

1 · WHO ARE YOU?

**Compliance office**
Operator · runs the programmeoperator

A

Ana
North depot · Driverenrolled

B

Ben
North depot · Warehouseenrolled

C

Cai
Head office · Managerenrolled

E

Eve
Not on staff · claims North depotnot enrolled

**Private state** on this device only

Operator secret3c02ad2c5b...061e

The chain only stores hash(operator secret). Operator actions prove knowledge of the secret.

2 · WHAT YOU CAN DO

Enrol staff
Each employee hands over a **commitment** = hash(secret, site, role). The operator learns nothing it can later match to a report.

Ana North depot · Driver
1687862e...1e517d✓ enrolled

Ben North depot · Warehouse
6f94cc55...dfc1b9✓ enrolled

Cai Head office · Manager
0a43cef5...6fa95c✓ enrolled

Rounds
Each member can file one report per round. Round 1 is open.

Open round 2

Case inbox
Report text arrives encrypted off-chain. Murmur checks it against the hash on the ledger, so it can't be altered.

#0 · Safety hazardseverity 5Investigating

"Bay 4 forklift brakes are failing. Supervisor said keep using it until month end."

✓ matches on-chain hash

From: North depot · role hidden · 2 corroborations

InvestigatingResolvedDismissed

3 · PUBLIC LEDGER WHAT EVERYONE SEES

✓ setStatus accepted

orgNameExample Freight Co. (synthetic)

operator6e64c70c...5f5c83 hash only

round1

rosterSize3

roster root711a562a...f2fc55

nullifiers34fac1a...5e9f 991105...9b34 0489b1...a2ff

reportCount1

reports[0]Investigating

categorySafety hazard

severity5

siteNorth depot

rolehidden

contentHash5ee508...2834

authorTag571801...31de

corroborations2

claimedByunclaimed

Nowhere on this ledger is a name, a roster position, or anything that links two reports to the same person.

Transactions

setStatus✓
chain sees: operator key ✓ (actually Compliance office; only this demo knows)
report #0 → Investigating
public: status

corroborate✗
chain sees: a prover (actually Ana; only this demo knows)
rejected by the contract: authors cannot corroborate their own report

corroborate✓
chain sees: an enrolled member (ZK proof) (actually Cai; only this demo knows)
report #0 corroborated

Personas · private state vs public ledger · per-transaction "public / kept private" log · guided story. No wallet, no network.

northstar-trustrail.github.io/murmur

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Tested adversarially

- ✓ only the operator can enrol, open rounds, set status
- ✓ discloses exactly the opted-in attributes
- ✓ can't lie about a disclosed attribute
- ✓ forged Merkle paths rejected inside the circuit
- ✓ outsiders can't file
- ✓ one report per member per round
- ✓ nothing links two reports by one member
- ✓ corroboration counted once per member
- ✓ authors can't corroborate themselves
- ✓ corroborating doesn't use up your report
- ✓ only the author can claim, and only once
- ✓ severity bounds enforced
- ✓ operator key hash only on chain
- ✓ **13 / 13 passing** (vitest on compact-runtime)

The "dishonest client" tests swap in malicious witness code, which shows the guarantees come from the circuit, not from honest clients.

Who pays, and why Midnight

- Employers with hotline duties: logistics, manufacturing, healthcare, retail (EU Whistleblower Directive, ethics-line rules)
- Unions, universities, member associations
- Per-seat SaaS operator console. The contract is the neutral core that neither vendor nor employer can quietly edit.

Why Midnight

Private state plus explicit `disclose()` make "prove membership, reveal nothing else" the default, not an add-on. Selective disclosure is a first-class language feature, and the public ledger gives auditors a tamper-evident case history.

This wave, and next

Built in Wave 2

Compact contract (6 circuits) · full ZK compile · TS witnesses + simulator · 13 tests incl. adversarial witnesses · browser demo on GitHub Pages · README, slides, video

Next

Preprod deployment with proof server + Lace DApp connector · encrypted off-chain text delivery · epoch-based roster rotation (revocation) · multi-party enrolment · k-anonymity warnings for small sites

Honest limits today: runs against the Compact runtime locally (not yet on a live network); report text delivery is simulated; the operator is trusted not to enrol fake members (roster size is public for auditing).

Murmur

Speak up. Stay unknown. Be believed.

Code (Apache-2.0): `github.com/northstar-trustrail/murmur`

Demo: `northstar-trustrail.github.io/murmur`